TEXORA · LECTURE SHEET

# Printing Processes and Special Printing Methods

*If dyeing colours the whole cloth, printing colours an argument on its surface. This sheet follows the print from design to washed fabric — paste, printing, fixation — and then the special methods that dyeing chemistry makes possible: discharge, resist, burnout, flock and foil.*

WP-4101 · TEXTILE CHEMISTRY & COLORATION-III · TOPIC 4 · SPECIALIZATION — WET PROCESSING

## 1. The printing chain

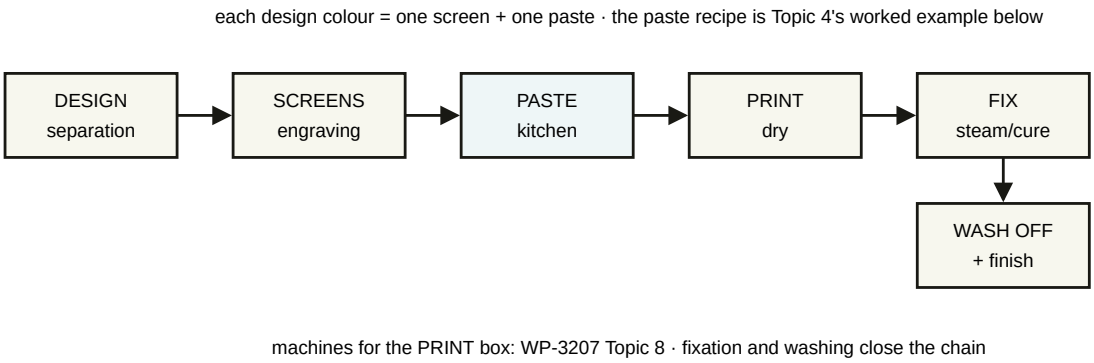



Fig. 1 — The printing chain. Printing is one box in a six-stage process; most print faults are born in the boxes before or after it.

Printing differs from dyeing in one essential: the colourant is applied in a **thickened paste** that must stay where it is put — no migration, no flushing into the ground — until fixation drives the dye from the paste film into the fibre locally. Thickener choice (alginate for reactive prints, because it does not react with the dye; emulsion and synthetic thickeners elsewhere) and paste rheology are therefore printing variables with no dyeing equivalent.

## 2. Print paste calculation

**WORKED EXAMPLE 1 — REACTIVE PRINT PASTE, PER 100 KG**

Recipe (parts by weight): dye 3.0 kg; urea 10.0 kg (humectant and dye solvent for steaming fixation); sodium alginate thickener stock (8% solids) 55.0 kg — contributing 4.4 kg of alginate

solids; soda ash 2.5 kg (alkali for fixation); resist salt (mild oxidant protecting the dye during steaming) 1.0 kg; water to weight.
Sum of named parts = 3.0 + 10.0 + 55.0 + 2.5 + 1.0 = 71.5 kg → water = 100 − 71.5 = **28.5 kg**.
Total = **100.0 kg ✓**.
Paste consumption planning: if the design covers the fabric at an average wet paste laydown of, typically, 120 g/m² on a 1.6 m wide fabric, one 100 kg paste batch prints 100,000 g / (120 × 1.6 g per metre) ≈ 520 m — the colour kitchen batches pastes against metres, exactly as dyeing batches against kilograms.

## 3. Styles of printing

- **Direct style:** the print paste carries the dye; colour appears where printed, fixed by steaming/curing. The standard route for reactive, pigment and disperse prints.
- **Discharge style:** the fabric is first dyed in a dischargeable ground dye; the print paste contains a reducing agent (e.g. sulfoxylate systems) that destroys the ground dye where printed, optionally with a discharge-resistant dye in the same paste (*illuminated discharge*) — white or coloured motifs on deep grounds with a softness no surface print matches.
- **Resist style:** the paste prevents dyeing where applied — mechanically (wax, in batik tradition; pastes that block print/dye penetration) or chemically (agents that prevent fixation under the print) — and the fabric is piece-dyed afterwards; the printed areas emerge white or in the resist colour.

## 4. Special printing methods

| METHOD | PRINCIPLE AND EFFECT |
|---|---|
| Burnout (devoré) | Print paste with acid (or acid-liberating salt) chars away one fibre of a blend on heating — classically polyester/cotton — leaving sheer patterned areas of the surviving fibre |
| Flock printing | Adhesive printed, short fibres (flock) applied and oriented electrostatically, fixed on curing — velvet-like raised motifs |
| Foil printing | Adhesive printed and cured with metallised foil pressed on — metallic gold/silver and holographic effects |
| Puff (expanded) print | Paste containing heat-expandable microspheres swells on curing into a raised 3D motif |
| Glitter print | Glitter particles suspended in a binder paste, printed and cured |
| Transfer (sublimation) | Design printed on paper with disperse dyes, then heat-pressed to polyester: dye sublimes from paper into fibre — no water, no wash-off; the polyester route for |

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|---|---|
| printing | photographic designs |
| Digital inkjet printing | Topic 8 (WP-3207): drop-on-demand inks on pre-treated fabric; reactive, acid, disperse and pigment ink sets; steaming and washing as in conventional fixation |

## 5. Fixation and washing-off — where prints are won

After printing and drying, reactive prints are fixed in saturated steam (about 102 °C, minutes-scale, by class), disperse prints on polyester by high-temperature steaming or thermosol, pigment prints by dry curing that polymerises the binder (150–160 °C class). Washing-off then removes thickener and unfixed dye: the white ground must emerge unstained, so washing is staged — cold rinse, warm rinse, hot soaping, final rinse — with the first baths carrying away most of the loose colour before it can re-deposit. A print that was perfect off the machine can be ruined in a single careless hot wash; the chain of Fig. 1 is only as strong as its last box.

## 6. Glossary — key terms

- **Laydown:** mass of wet paste applied per unit area of fabric.
- **Illuminated discharge:** discharge printing in which a resistant dye in the paste leaves a coloured motif.
- **Devoré:** burnout printing destroying one component of a blend.
- **Resist salt:** a mild oxidising agent in reactive print pastes protecting dyes during steaming.
- **Thermosol:** dry-heat fixation of disperse dyes on polyester.

## 7. Key points

- The paste is a recipe by weight that must sum exactly (71.5 kg of ingredients + 28.5 kg water = 100 kg) and lay down predictably per square metre.
- Discharge destroys colour, resist prevents it, burnout removes fibre — three special effects that are all chemistry scheduled in the print paste.
- Fixation and washing-off close the chain; ground staining in washing is the commonest way good printing is lost.

## 8. Summary

Printing processes combine the whole course: machines from WP-3207, dye chemistry from Topic 2, process discipline from Topic 3. The special methods extend the same logic — every effect on a printed

fabric is a paste that does something local: fixes, destroys, resists, dissolves, glues or expands. Learn the paste, and the effect follows.

## 9. Practice

1. A pigment print paste is made per 100 kg: pigment 6 kg, binder 18 kg, thickener stock 60 kg, fixer 1.5 kg, softener 2 kg, water to weight. Calculate the water required.
2. Why is sodium alginate specified for reactive print pastes rather than a starch thickener?
3. A design needs white flowers on a deep navy cotton ground, with a hand as soft as the dyed fabric itself. Choose a printing style and justify it against direct pigment printing.
4. State the fibre pair and the chemical action in a classic burnout print, and the safety check its paste demands.

### Answer key

1. Named parts = 6 + 18 + 60 + 1.5 + 2 = 87.5 kg → water = 100 − 87.5 = 12.5 kg.
2. Alginate does not react with reactive dyes (its carboxyl groups are ionised and unreactive toward the dye under printing conditions), so the dye remains free to fix on the fibre and the thickener washes out cleanly. Starch carries hydroxyls that react with the dye like cellulose does, wasting dye in the paste film and dulling the print.
3. Discharge style. The white is produced by destroying the ground dye within the fabric, so there is no surface film: handle stays that of dyed cloth, unlike a white pigment print, which sits on the surface and stiffens the motif. (Requires a dischargeable navy dye — specified at the dyeing stage.)
4. Polyester/cotton blend; the printed acid carbonises and removes the cotton on heating, leaving polyester lace-like areas. Safety check: acid strength and cure conditions must fully destroy the target fibre without attacking the surviving polyester or creating excessive fumes/strength loss at the motif edges.

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Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib